

$$E\left(\sum_{i=1}^N X_i\right)$$

$$= E_N\left[E\left(\sum_{i=1}^N X_i \mid N\right)\right]$$

$$= E_N(N E(X_i))$$

$$= E_N(N\theta)$$

$$= \theta E_N(N)$$

$$= \theta \sum_{n=0}^{\infty} n e^{-\lambda} \frac{\lambda^n}{n!}$$

$$= \theta \lambda \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^{n-1}}{(n-1)!}$$

$$n^* = n-1$$

$$= \theta \lambda \sum_{n^*=0}^{\infty} \frac{e^{-\lambda} \lambda^{n^*}}{n^*!}$$

$$= \theta \lambda e^{-\lambda} e^{\lambda}$$

$$= \theta \lambda$$